

PIN-Lite: Efficient and Explainable Multimodal Deepfake Detection via Knowledge Distillation

1st Author Name
Dept. of Computer Science
University/Institute Name
City, Country
email@example.com

2nd Author Name
Dept. of Computer Science
University/Institute Name
City, Country
email@example.com

Abstract—The proliferation of hyper-realistic deepfakes necessitates robust and deployable detection systems. While current state-of-the-art multimodal models like PinPoint achieve high accuracy by leveraging audio-visual inconsistencies, their excessive computational cost limits deployment on edge devices. Furthermore, compressing these models often compromises their interpretability, a critical factor for trustworthy AI. To address this, we propose PIN-Lite, a lightweight multimodal deepfake detector. We introduce a comprehensive compression pipeline combining Knowledge Distillation (KD) and Structured Pruning, designed to preserve not just the class predictions but also the internal reasoning of the heavy teacher model. We also propose a novel metric, the Explainability Preservation Score (EPS), to quantify the alignment of saliency maps between teacher and student.

Our experiments on the LAV-DF dataset demonstrate that PIN-Lite reduces the model size by approximately $8.5\times$ (from 57.32 MB to 6.62 MB) and improves inference latency by $2.2\times$ (from 82.5 ms to 37.6 ms) while matching or even slightly exceeding the teacher’s accuracy (97.53% vs. 97.37%). We further show that simple quantization of the teacher model catastrophically degrades performance (52.13% accuracy), validating our distillation-based approach. PIN-Lite represents a significant step towards efficient, explainable, and trustworthy deepfake detection at the edge.

Index Terms—Deepfake Detection, Knowledge Distillation, Model Compression, Explainable AI (XAI), Edge AI, Multimodal Learning

I. INTRODUCTION

The democratization of generative AI has led to a surge in sophisticated deepfakes, posing significant threats to privacy, security, and information integrity. Detecting these manipulations requires analyzing subtle inconsistencies across modalities (e.g., lip-sync errors between audio and video). State-of-the-art (SOTA) models typically employ complex architectures, such as dual-stream networks with heavy Transformers or ResNet backbones [1], [4], to capture these multimodal artifacts.

While effective, these models suffer from high latency and large memory footprints, making them unsuitable for real-time applications or deployment on resource-constrained edge devices (e.g., mobile phones, IoT cameras). Moreover, as AI systems are increasingly deployed in sensitive domains, *interpretability* becomes paramount. Users need to understand

why a video is flagged as fake. Standard model compression techniques like quantization or pruning often focus solely on maintaining accuracy, potentially altering the model’s decision boundary in opaque ways and destroying the “reasoning” capability found in the original model.

To bridge this gap, we introduce **PIN-Lite**, a framework for creating efficient yet explainable deepfake detectors. Our approach leverages **Knowledge Distillation (KD)** not only to transfer predictive power but also to align the attention mechanisms of a lightweight “Student” model with a heavy “Teacher” model. This ensures that the student detects fakes for the “right reasons” (e.g., focusing on the mouth region for lip-sync errors).

Our key contributions are:

- 1) **PIN-Lite Architecture**: A lightweight multimodal student network that achieves SOTA accuracy (97.53%) with significantly reduced computational cost (6.62 MB size, 37.6 ms latency).
- 2) **Explainability Preservation Score (EPS)**: A novel metric combining Spearman correlation and Intersection over Union (IoU) of saliency maps to quantitatively evaluate how well a compressed model retains the teacher’s interpretability.
- 3) **Comprehensive Ablation**: A rigorous comparison of Distillation, Structured Pruning, and Direct Quantization, highlighting the failure of naive quantization for complex multimodal teachers.

II. RELATED WORK

A. Multimodal Deepfake Detection

Recent advancements focus on exploiting inconsistencies between video and audio streams. M2TR [1] utilizes a multi-scale transformer to detect local inconsistencies and frequency artifacts, achieving high robustness against compression. Similarly, CAD [4] employs cross-modal alignment and distillation, leveraging large pre-trained encoders like CLIP and Whisper to detect semantic mismatches. While highly accurate, these models rely on heavy backbones (EfficientNet-B4, CLIP-ViT), resulting in high computational costs unsuitable for edge deployment. AV-Lip-Sync+ [3] further pushes accuracy (99.29% on FakeAVCeleb) using AV-HuBERT and ViViT but

with a massive parameter count ($\sim 250\text{M}$), reinforcing the need for efficient alternatives.

B. Model Compression and Distillation

Knowledge Distillation (KD) is a proven technique for transferring knowledge from a large teacher to a smaller student. Karathanasis et al. [7] demonstrated that up to 90% compression is possible in deepfake detection using KD without significant accuracy loss on in-domain data. Beyond standard logit-based distillation, recent works have focused on transferring interpretability. XDistillation [5] uses a convolutional autoencoder to distill "explanations" alongside predictions. Similarly, e²KD [6] forces the student's gradient-based explanations to match the teacher's via cosine similarity loss. Our work builds on these concepts but specifically targets the multimodal nature of deepfake detection, ensuring audio-visual attention alignment.

C. Explainable AI (XAI) in Forensics

Explainability is crucial for user trust. DF-P2E [2] proposes a completely different approach, generating natural language explanations using Large Language Models (LLMs) like LLaMA-3. However, this introduces extreme latency ($\sim 28\text{s}$ per image), making it impractical for real-time use. PIN-Lite aims to retain the visual explainability (saliency maps) of complex models while running at millisecond-level latency.

III. METHODOLOGY

A. The Teacher Model: PinPoint

We utilize the PinPoint model as our Teacher (T). It follows a standard multimodal architecture:

- **Video Encoder:** A ResNet-18 backbone extracts per-frame visual features.
- **Audio Encoder:** A 1D-CNN processes MFCC features from the audio stream.
- **Fusion Module:** A Transformer-based fusion layer with cross-attention mechanisms aligns audio and visual features to detect synchronization anomalies.

This model serves as the ground truth for both classification labels and attention maps during the distillation process.

B. The Student Model: PIN-Lite

The Student model (S) retains the structural design of the Teacher but with significantly reduced capacity:

- **Reduced Dimensions:** The embedding dimension is reduced (e.g., $256 \rightarrow 128$).
- **Fewer Heads:** The number of attention heads in the transformer layers is halved.
- **Lighter Backbone:** We employ a streamlined version of the video encoder.

C. Distillation & Pruning Strategy

We employ a multi-stage compression pipeline:

1) *Knowledge Distillation:* We train the student using a composite loss function:

$$L_{total} = \alpha L_{hard} + (1 - \alpha) L_{soft} + \beta L_{attn} \quad (1)$$

where L_{hard} is the standard cross-entropy loss against ground truth labels, and L_{soft} is the KL-divergence between the teacher's and student's softened logits. Crucially, L_{attn} minimizes the Mean Squared Error (MSE) between the teacher's and student's attention maps, forcing the student to focus on the same spatial-temporal regions (e.g., lips) as the teacher.

2) *Structured Pruning:* Post-distillation, we apply iterative structured pruning to the student. We identify and prune entire attention heads that contribute least to the L1-norm of the weights. This unstructured pruning is followed by fine-tuning to recover accuracy.

3) *Quantization (Teacher Baseline):* For comparison, we also explore direct quantization of the original Teacher model to INT8 using Quantization-Aware Training (QAT).

D. Explainability Preservation Score (EPS)

To quantitatively measure interpretability, we define the EPS. For a given input, let S_T and S_S be the flattened saliency maps of the teacher and student, respectively.

$$EPS = 0.5 \cdot \rho(S_T, S_S) + 0.5 \cdot \text{IoU}(\text{TOP}_{20\%}(S_T), \text{TOP}_{20\%}(S_S)) \quad (2)$$

where ρ is the Spearman rank correlation, and IoU measures the overlap of the top 20% most salient pixels. A high EPS indicates that the student has preserved the teacher's reasoning.

IV. EXPERIMENTAL SETUP

We evaluate our approach on the **LAV-DF** (Localized Audio-Visual DeepFake) dataset, which contains diverse manipulation types including lip-sync errors and visual artifacts.

- **Baselines:** Original PinPoint (Teacher), Pruned Student, Quantized Teacher.
- **Metrics:** Accuracy, F1-Score, AUC, Inference Latency (ms/sample on CPU), Model Size (MB), and EPS.

V. RESULTS AND ANALYSIS

A. Performance Comparison

Table I presents the comprehensive results. The **Distilled** PIN-Lite model achieves the best balance of performance and efficiency. Remarkably, it slightly outperforms the Teacher in accuracy (97.53% vs 97.37%), likely due to the regularization effect of distillation which prevents overfitting.

TABLE I
COMPARATIVE RESULTS OF MODEL VARIANTS

Model	Acc (%)	F1	AUC	Lat (ms)	Size (MB)	EPS
Teacher (Base)	97.37	0.9821	0.9683	82.50	57.32	1.0
Distilled	97.53	0.9834	0.9584	37.60	6.62	0.60
Pruned	97.38	0.9824	0.9558	39.41	6.62	0.58
Quantized (T)	52.13	0.5273	0.6621	424.34	25.25	N/A

The **Quantized Teacher** performs poorly, with accuracy dropping to 52.13%. This suggests that the complex interactions in the Teacher’s fusion layers are highly sensitive to reduced precision without extensive architectural adaptation, validating the necessity of the ”Distill-then-Prune” approach over simple quantization.

B. Efficiency Analysis

PIN-Lite (Distilled) reduces the model size by $\sim 8.5\times$ and latency by $\sim 2.2\times$. This makes it suitable for real-time detection on standard hardware. As shown in the Pareto analysis (represented as ”Figure 1” in our supplementary material), PIN-Lite occupies the optimal top-left corner of the Accuracy vs. Latency frontier.

C. Explainability Preservation

While compression often degrades interpretability, the Distilled model maintains an EPS of 0.60. This indicates a substantial alignment with the Teacher’s attention, far better than random chance. The Pruned model follows closely with an EPS of 0.58. This preservation of reasoning ensures that PIN-Lite remains trustworthy despite its reduced size.

VI. CONCLUSION

We presented PIN-Lite, a specialized distillation framework for multimodal deepfake detection. By distilling knowledge and attention maps from a heavy Teacher, we created a model that is $8.5\times$ smaller and $2.2\times$ faster while maintaining SOTA accuracy. The failure of direct quantization highlights the value of our specialized pipeline. With high accuracy, low latency, and preserved explainability (EPS), PIN-Lite is a robust candidate for edge-based deepfake defense.

REFERENCES

- [1] J. Wang et al., ”M2TR: Multi-modal Multi-scale Transformers for Deepfake Detection,” in *ICMR ’22*, 2022.
- [2] S. Tariq et al., ”From Prediction to Explanation: Multimodal, Explainable, and Interactive Deepfake Detection Framework,” in *ACM MM ’25*, 2025.
- [3] S. A. Shahzad et al., ”AV-Lip-Sync+: Leveraging AV-HuBERT to Exploit Multimodal Inconsistency,” *IEEE THMS*, 2025.
- [4] Y. Du et al., ”CAD: A General Multimodal Framework for Video Deepfake Detection via Cross-Modal Alignment and Distillation,” *arXiv preprint*, 2025.
- [5] R. Alharbi et al., ”Learning Interpretation with Explainable Knowledge Distillation,” in *IEEE Big Data*, 2021.
- [6] A. Parchami-Araghi et al., ”Good Teachers Explain: Explanation-Enhanced Knowledge Distillation,” *CVPR*, 2022.
- [7] A. Karathanasis et al., ”A Brief Review for Compression and Transfer Learning Techniques in DeepFake Detection,” *arXiv*, 2025.